

1. Prerequisites
2. Create S3 Buckets

Amazon S3 > Buckets

Buckets

General purpose buckets | All AWS Regions | Directory buckets

General purpose buckets (4) Info

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	Creation date
aws-glue-assets-804792415385-us-east-1	US East (N. Virginia) us-east-1	April 14, 2026, 11:31:11 (UTC-04:00)
handson-11-cloudcomputing	US East (N. Virginia) us-east-1	April 7, 2026, 11:27:46 (UTC-04:00)
handson12finallanding	US East (N. Virginia) us-east-1	April 14, 2026, 11:24:07 (UTC-04:00)
handson12finalprocessed	US East (N. Virginia) us-east-1	April 14, 2026, 11:25:36 (UTC-04:00)

3. Create IAM Role for AWS Glue

IAM > Roles

Identity and Access Management (IAM)

Search IAM

- Dashboard
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 - Access Analyzer
 - Resource analysis
 - Unused access
 - Analyzer settings
 - Credential report
 - Organization activity
 - Service control policies
 - Resource control policies
- IAM Identity Center
- AWS Organizations

Roles (6)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Search

Role name	Trusted entities	Last activity
AWSGlueServiceRole-Reviews	AWS Service: glue	10 minutes ago
AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2 (S	18 minutes ago
AWSServiceRoleForSupport	AWS Service: support (Service-Link	-
AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service	-
Handson11GlueRole	AWS Service: glue	7 days ago
start_glue_job_trigger-role-4py1psil	AWS Service: lambda	12 minutes ago

Roles Anywhere

Authenticate your non AWS workloads and securely provide access to AWS services.

Access AWS from your non AWS workloads

Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.

X.509 Standard

Use your own existing PKI infrastructure or use [AWS Certificate Manage Authority](#) to authenticate identities.

4. Create the AWS Glue ETL Job

The screenshot displays the AWS Glue console interface. On the left, a navigation pane lists various services like 'Getting started', 'ETL jobs', 'Data Catalog', and 'Data Integration and ETL'. The main area shows the configuration for a job named 'process_reviews_job'. A green banner at the top indicates 'Successfully updated job'. Below this, tabs for 'Script', 'Job details', 'Runs', 'Data quality', 'Schedules', and 'Version Control' are visible. The 'Script' tab is active, showing a Python script for an ETL job. The script imports necessary modules, initializes contexts, reads data from S3, and performs transformations. A success message at the top of the console reads: 'Successfully updated job process_reviews_job. To run the job choose the Run Job button.'

```
1 import sys
2 from aws glue.transforms import *
3 from aws glue.utils import getResolvedOptions
4 from pyspark.context import SparkContext
5 from aws glue.context import GlueContext
6 from aws glue.job import Job
7 from pyspark.sql.functions import col, to_date, upper, coalesce, lit
8 from aws glue.dynamicframe import DynamicFrame
9
10 ## Initialize contexts
11 sc = SparkContext()
12 glueContext = GlueContext(sc)
13 spark = glueContext.spark_session
14 job = Job(glueContext)
15
16 # --- Define S3 Paths (Updated with your new names) ---
17 s3_input_path = "s3://handson12finallanding/"
18 s3_processed_path = "s3://handson12finalprocessed/processed-data/"
19 s3_analytics_path = "s3://handson12finalprocessed/Athena Results/"
20
21 # --- Read the data from the S3 landing zone ---
22 dynamic_frame = glueContext.create_dynamic_frame.from_options(
23     connection_type="s3",
24     connection_options={"paths": [s3_input_path], "recurse": True},
25     format="csv",
26     format_options={"withHeader": True, "inferSchema": True},
27 )
28
29 # Convert to a standard Spark DataFrame for easier transformation
30 df = dynamic_frame.toDF()
31
32 # --- Perform Transformations ---
33 # 1. Cast 'rating' to integer and fill null values with 0
34 df_transformed = df.withColumn("rating", coalesce(col("rating").cast("integer"), lit(0)))
35
36 # 2. Convert 'review date' string to a proper date type
```

5. Create the Lambda Trigger Function

The screenshot shows the AWS Lambda console for a function named 'start_glue_job_trigger'. A green banner at the top states 'Successfully updated the function "start_glue_job_trigger"'. The function is configured with an S3 trigger. The 'Code source' tab is selected, displaying a Python script. The script defines a lambda handler that starts a Glue ETL job when triggered by an S3 event. The script includes comments and error handling. The console also shows the function's ARN, description, and last modified time. The code source is as follows:

```
1 # Replace with the name of your Glue job
2 GLUE_JOB_NAME = "process_reviews_job"
3
4 def lambda_handler(event, context):
5     """
6     This Lambda function is triggered by an S3 event and starts a Glue ETL job.
7     """
8     glue_client = boto3.client('glue')
9
10    try:
11        print(f"Starting AWS Glue job: {GLUE_JOB_NAME}")
12        response = glue_client.start_job_run(JobName=GLUE_JOB_NAME)
13        print(f"Successfully started job run. Run ID: {response['JobRunId']}")
14        return {
15            'statusCode': 200,
16            'body': f"Glue job {GLUE_JOB_NAME} started successfully."
17        }
18    except Exception as e:
19        print(f"Error starting Glue job: {e}")
20        raise e
```

Outputs:

Amazon S3 > Buckets > handson12finalprocessed > Athena Results/

Athena Results/ [Copy S3 URI](#)

Objects Properties

Objects (5) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	run-1776184182990-part-r-00000	-	April 14, 2026, 12:29:48 (UTC-04:00)	567.0 B	Standard
☐	run-1776185054468-part-r-00000	-	April 14, 2026, 12:44:18 (UTC-04:00)	567.0 B	Standard
☐	run-1776185059639-part-r-00000	-	April 14, 2026, 12:44:22 (UTC-04:00)	1003.0 B	Standard
☐	run-1776185062259-part-r-00000	-	April 14, 2026, 12:44:27 (UTC-04:00)	66.0 B	Standard
☐	run-1776185067552-part-r-00000	-	April 14, 2026, 12:44:28 (UTC-04:00)	48.0 B	Standard

Amazon S3 > Buckets > handson12finalprocessed > processed-data/

processed-data/ [Copy S3 URI](#)

Objects Properties

Objects (2) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

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Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	run-1776184173758-part-r-00000	-	April 14, 2026, 12:29:39 (UTC-04:00)	10.5 KB	Standard
☐	run-1776185041550-part-r-00000	-	April 14, 2026, 12:44:07 (UTC-04:00)	10.5 KB	Standard